

Executive Summary:

Cross-Lingual Refusal Under English-Only Abliteration

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1 Goal

Measure whether Slovene-taught refusal in GaMS3-12B-Instruct transfers to English, and whether an English-only de-censoring edit (Heretic ablation) removes Slovene refusal as effectively as English refusal, compared with Gemma-3-12B-IT. The two models share the same pretrained ancestor but diverge in safety-supervision language. The practical stake is whether a monolingual adversary can strip bilingual safety, relevant for evaluating the robustness of bilingual safety training.

2 What Was Tried

Five iterations ran 24 artifacts across 17 experiments, 5 evaluations, and 2 research checks. Designs included: (i) difference-in-differences on RefuseEU prompts across hazard categories; (ii) signal-detection decomposition (sensitivity d' and criterion c) on HARD items; (iii) Heretic English-objective ablation with transfer curves, lambda-scaling ladders, and fresh-trial confirmation; (iv) a $2 \times 3 \times 3$ input-language $\times$ output-language $\times$ dose mechanism test; (v) teacher-inheritance screening; (vi) base-checkpoint geometry and persona-gate tests; (vii) utility controls (Belebele, ARC-Challenge, BoolQ); (viii) R-INCAP safety-vs-production-failure decomposition with StrongREJECT harmful-content scoring.

3 Key Results

- **C-OUT confirmed.** Gemma’s residual refusal follows the output language: output-language contrast (PPI-corrected) = $+2.43$ [$+0.89$, $+5.38$] log-odds (experiment 16). Cross-model difference = -2.31 [-5.34 , -0.68]: GaMS3 shows no output-language effect. The input-language effect in Gemma is negative (-0.68 [-1.23 , -0.36]).
- **R-BASE, not edit-induced.** The lag is pre-edit baseline, not created by the edit. Experiment 15 (most powerful estimate): $G_{3,\text{edit}} = +0.47$ [-0.96 , 2.14], spanning zero. Parallel-curve geometry holds across bodies.
- **Reserve is safety, not production failure.** Harmful content separates at both doses ($\text{OUT}_U = 2.61$ [2.20 , 3.07] at high dose). Production-failure index difference: Gemma 0.46 [0.23 , 0.65]; half of non-compliance is deflection, consistent with the relevance curve.
- **Criterion shift replicates.** Gemma’s criterion moves toward refusal on Slovene-output items (8 of 9 compliance-valid rows have CIs below zero). GaMS3’s criterion is language-invariant.

- **Edit is effective everywhere.** On RefusEU FINAL ($n = 1,300$ per cell), all four model $\times$ language cells are ROBUST.
- **Utility preserved.** No Slovene-specific downstream damage; localised ARC-Challenge cost of 3.7 pp for GaMS3.

4 What Failed

- **Alternate 2 (base-checkpoint geometry):** Dead. Base-model separability does not predict instruct-level refusal.
- **Alternate 3 (persona gating):** Untestable. Manipulation check failed; ablating the own-name direction substituted the Qwen teacher identity.
- **Alternate 4 (prefill depth):** Uncitable. No readout reached kappa 0.60 on prefill continuations.
- **Ancestor-direction induction (C4):** Failure branch. The pretrained direction never induced refusal above 0.50.
- **Readout gate:** Never passed across five iterations. Best edited-row kappa was 0.43 (gemini-2.5-flash).
- **Experiment 13:** GPU lost before execution.
- **Dose contrast (D):** Underpowered in every iteration ($\text{MDE} > 2m$).

5 Best Result

The most powerful lag estimate is experiment 15’s $G_{3,R} = -0.70 [-1.01, -0.42]$ (MDE 0.42), surviving under tipping-point-adjusted (TTJ) readout ($-0.79 [-1.09, -0.48]$). The best-supported mechanistic account is C-OUT (output-language effect): English-only ablation amplifies Gemma’s Slovene–English refusal gap rather than closing it, and the residual is real safety (harmful content separates). The narrowed novelty claim (crossing both prompt and response language with safety endpoints on a sibling pair under partial-dose ablation) is unoccupied as of September 2026 (medium-low confidence), although the broader output-language finding is partially occupied by Addagada and Nguyen et al.

6 Open Questions

- The readout gate has never passed. All adjudication is author-LLM, not human.
- The frozen-core Heretic procedure (bf16, 80 trials) has never been run.
- No artifact has scored natural RefusEU reserved-split prompts under the edit.
- The dose contrast remains underpowered: the main claim vs supervised-reach rivalry is unresolved.
- Novelty is PARTIALLY OCCUPIED and may narrow further.

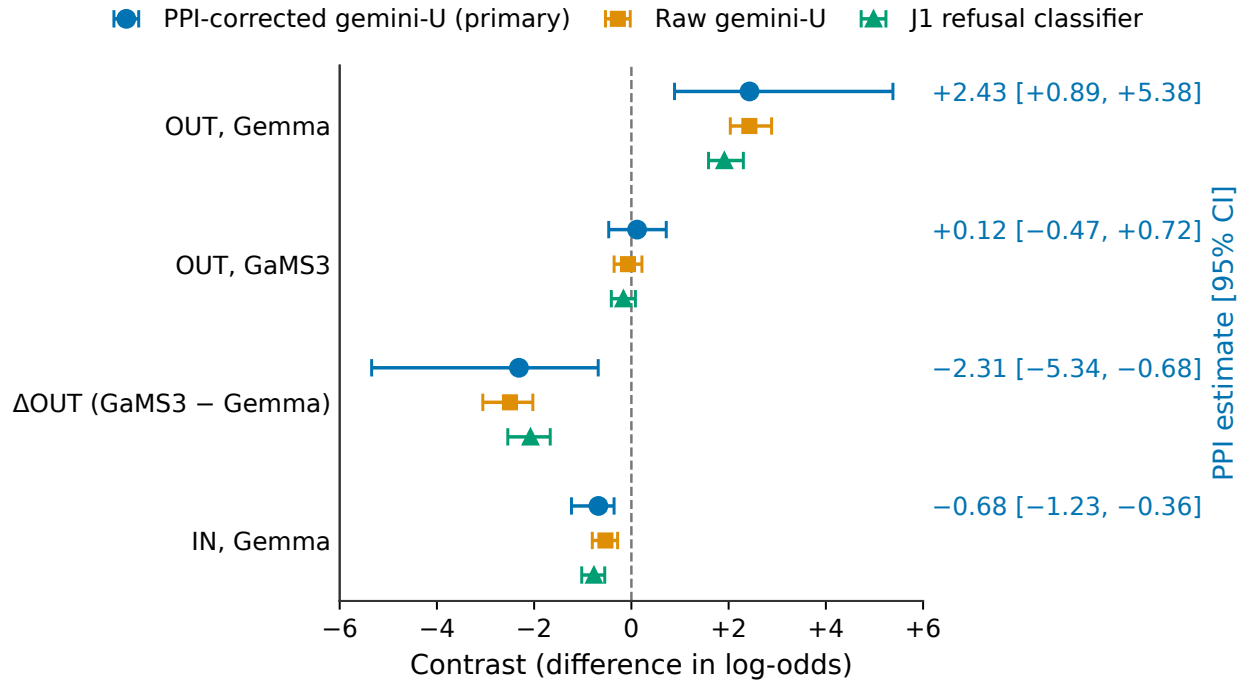


Figure 1: Output-language contrasts on the held-out 294-item confirmation set (Experiment 16). Gemma’s output-language contrast is +2.43 [+0.89, +5.38] log-odds (PPI-corrected). GaMS3 shows no output-language effect (+0.12 [-0.47, +0.72]). Cross-model difference: -2.31 [-5.34, -0.68].